On the Performance of Multihop Cognitive Wireless Powered D2D Communications in WSNs

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***Abstract*—This paper studies the performance of multi-hop cognitive wireless powered device-to-device (D2D) communications in wireless sensor networks (WSNs). In our analysis, each sensor node harvests energy from multiple dedicated power beacons and shares the spectrum resources with multiple primary receivers (PRs) using underlay cognitive radio. Additionally, we consider a practical scenario of cognitive wireless powered D2D communications in WSNs, where the knowledge of interference channels is assumed to be imperfect. To improve the network performance, we propose two user scheduling schemes, namely dual-hop scheduling (DHS) and best-path scheduling (BPS) schemes. We then investigate the performance of the proposed scheduling schemes in terms of outage probability and outage floor. Through numerical results, we show that BPS scheme significantly outperforms DHS scheme, which in turns outperforms the state-of-the-art solution. Moreover, the advantages and drawbacks of each scheme are analyzed and discussed comprehensively. We also point out that the inaccurate knowledge of the interference channels significantly affects any performance metric of multi-hop cognitive D2D communications in WSNs such as outage probability, outage floor, and performance loss.**

***Index Terms*—Cognitive radio, multihop device-to-device (D2D) communication, outage probability, user scheduling, wireless sensor networks, wireless powered.**

I. Introduction

Recently, wireless sensor networks (WSNs) have envisioned as a networking solution to collect and efficiently monitor the environment conditions in the coming Internet-of-Thing (IoT) era [1]. The explosion of unlicensed spectrum devices makes the traditional WSNs to undergo severe interference caused by other applications using the same spectrum [2]. Cognitive radio (CR) for WSNs is developed as a spectrum-efficient technology to reduce such interference and improve spectral utilization [3]. Therefore, various applications are evolving with IoT services and expected to be more efficient for cognitive radio WSNs in the near future.

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Device-to-device (D2D) communication in 5G and beyond provides an effective infrastructure, where WSNs are integrated with 5G networks using D2D communications or ad hoc links to activate specific applications such as smart cities, IoT applications, and public safety networks [24]. The benefits of D2D communications are brought by enabling sensor devices to directly communicate with one another in proximity without significant infrastructure [27]. Exploiting proximity transmission brings a wide range of benefits such as high data rate, low latency, and reliability [26]. As a result, D2D communication is deployed as an important feature for public safety networks, IoT ecosystems and WSNs [27].

In large-scale WSNs, multi-hop transmission is a feasible solution to extend the radio coverage and provide reliable communications [27]. However, one of the main challenges in wireless sensor networks is how to prolong the network lifetime and maintain the connectivity since the sensor nodes are generally small devices with limited battery capacity. Wireless power transfer (WPT) technique has been deployed in WSNs to efficiently solve such a problem because each sensor device can harvest energy from dedicated power beacons (PBs) to support its operation [28]. Thus, this paper focuses on the multi-hop cognitive wireless powered D2D communications in WSNs, where we aim to understand the impacts of related parameters (e.g., number of PBs, number of sensor nodes, number of primary users, etc) on the system performance and give insightful information for network planning.

A. Related Works and Motivations

In this subsection, we discuss the most recent achievements of WPT in IoT ecosystems and WSNs. The recent advances in microelectronic devices have ensured that energy harvesting (EH) circuits are capable of harvesting micro-watts to milliwatts of power over the range of several kilometers, as in some practical experiments (see, e.g., [29]–[31]). Radio frequency (RF) signal-based WPT was capable of transferring energy over a long distance to activate sensor nodes and maintain their operation [29]. In [30], the sensor nodes was able to harvest energy reliably from a broadcasting tower over long distance, and the dedicated power beacons successfully delivered the energy to sensors and smart phones to support their operation [31]. With these achievements, WPT technique provides an efficient mean to support limit-battery devices in IoT ecosystems and WSNs.

Recently, D2D communication underlaying cellular networks has emerged as one of the most widely studied paradigms [4]–[7], where mobile devices can enable physical-

proximity communication to save energy and improve the spectral efficiency. In [4], a D2D communication-based load balancing algorithm was introduced to efficiently achieve offload traffic among different tier cells according to real-time traffic distributions. It was proved in [5] that the quality of experience was inversely proportional to the dense user distribution in the networks. A D2D communication enabling multi-channel cellular networks was investigated in [6], where a mobile user can select its transmission mode according to its received signal strength from the nearest base station (BS). Energy harvesting-based D2D communication has also received plenty of attention [8]–[20]. In [8], the authors considered EH-based D2D communication for large-scale cellular networks, where the power outage probability (OP) was improved with increasing density of PBs. Sum-rate of D2D links in heterogeneous networks was maximized by appropriately designing spectrum sharing strategies [9]. On the other hand, in [10] the energy efficiency (EE) was improved by deploying WPT enabled D2D communication. To achieve a good balance between the traffic offloading gains and the WPT costs, the authors in [11] proposed data-energy beamforming designs with EH-enabled D2D communication. In [12], the authors introduced the meta distribution to attain an efficient operating regime for wireless energy and information transfer in D2D networks. In [13], the average transmit power at D2D devices and system EE were improved by increasing the size of antenna array at the BS. Moreover, data rate and ergodic rate of D2D communication were maximized by designing the time duration in the EH phase [14]. The authors in [15] studied methods to prolong the network lifetime and reduce power consumption of D2D devices while fulfilling the quality of service (QoS) requirements. In [16], system EE and EH at users were achieved by applying stable matching algorithm for D2D pairs and cellular users. The authors in [17] showed that the power OP of cellular network was reduced proportionally with either the number of antennas or the transmit power of PB. Various optimal spectrum access policies were considered in [18] to improve the OP of signal-to-interference-plus-noise ratio for D2D users as well as mobile users. In [19], the authors solved the optimization problem of maximum sum-rate subject to QoS of cellular and D2D links, energy harvesting, and sub-carrier power constraints. It was also shown in [20] that the OP of the heterogeneous network was smaller than that of its counterpart without EH-based D2D communications.

EH-based multi-hop communications play a vital role in extending transmission range and network coverage. A multi-hop cooperative cluster-based network was studied in [32] to improve the system OP. Some recent advances in simultaneous wireless information and power transfer (SWIPT) systems were surveyed in [21], where a novel SWIPT-enabled D2D communication was realized in cellular networks to improve the power and system performance. In [7], a general network coding scenario enabling multi-hop D2D communications was proposed to improve data transmission efficiency in cellular networks supporting multimedia contents. A novel EH-based multi-hop cognitive radio network (CRN) was proposed in [36], where each relay harvests energy from co-channel interferences for its operation. Recently, various path-selection approaches in EH multi-hop WSNs were studied in [37]. Xu *et al.* in [39] proposed an EH multi-hop CRNs with single relay, a PB and a primary receiver (PR). However, it was considered a limited scenario in which impractical condition of interference channels is assumed for performance analysis.

All the aforementioned works mainly focused on a multi-hop transmission with single relay or perfect channel state information (CSI) of interference links [32], [36]–[39], which is highly idealistic. In practice, sensor nodes are densely deployed and grouped into clusters. Selecting the best cooperative sensor node in each cluster for D2D communication will provide a more reliable and efficient transmission [38]. Moreover, the uncertainties of interference channels are inevitable due to channel estimation error, mobility or feedback delay [40]. Therefore, in this work we aim to consider a general multi-hop cognitive wireless powered D2D communications for multiple PRs in WSNs with practical conditions of imperfect CSI. Furthermore, we exploit relay selection criteria for multi-hop network for enhancing communication reliability and spectrum utilization[1]. Addressing these open research issues will give us an interesting insight into the system characteristics and unveil how the imperfect CSI impacts the network performance.

In the spectrum perspective, D2D communication is applied to WSNs to achieve efficient spatial reuse of the precious wireless band resources. That is one of the novelty features in our considered network and it is not available in traditional WSNs. Particularly, D2D nodes can operate simultaneously the licensed spectrum band to the 5G cellular networks using underlay D2D communications to improve the spectrum utilization. It was proved in [4], [22], [25] that underlay D2D communication was able to provide high spectrum efficiency, high power efficiency, and support spectrum sharing in the 5G networks and IoT ecosystems. Thus, we can expect that applying underlay D2D communication in WSNs will bring the highest possible spectrum efficiency in the future WSNs and IoT ecosystems. In the protocol perspective, we deploy multiple D2D clusters and adopt user scheduling schemes for cooperative WSNs, which have not considered in the traditional WSNs. Particularly, two user scheduling schemes, i.e., DHS and BPS, rely on the instantaneous signal-to-noise ratio (SNR) of received signals to establish the best cooperative multi-hop D2D communication route, which helps to improve the performance of multi-hop D2D communication even when undergoing interference constraint. The time division multiple access (TDMA) scheduling in media access control (MAC) layer is applied to the considered system setup [22], [23], which allows any D2D node transmits its data at a given sub-time slot and turns to idle mode on other sub-time slots to save energy, to avoid collision, and to reduce network delays [23]. In networking perspective, the sensor nodes are generally small with limited power capacity deployed in hazardous environments. It is impossible to replace or recharge the battery sensor nodes, which is one of the most critical unsolved problems in traditional WSNs. Therefore, we adopt the WPT-based

[1]When multiple relays in a cluster simultaneously transmit or receive signals, a number of available spectra will be occupied. Thus, employing relay selection method will alleviate this problem.

D2D communication in our considered WSN to tackle such energy-constraint problem. Moreover, D2D communication can support high-speed data rate, e.g., it achieves maximum data rate of 1 Gbps in some realistic applications such as public safety, content sharing, local advertising, cellular relay [26]. Thus, we expect that deploying D2D communication in WSNs will achieve the highest possible data rate in the future cognitive WSNs.

B. Contributions and Organization

In this paper, we study the impacts of cooperative sensor node selection on the performance of multi-hop cognitive wireless powered D2D communications in WSNs. Specifically, we propose two user scheduling schemes for the considered system to improve the end-to-end system performance, namely dual-hop scheduling (DHS) and best-path scheduling (BPS) schemes. Particularly, the DHS scheme tries to select the two best cooperative sensor nodes in two consecutive clusters to form a dual-hop transmission while the BPS scheme selects the most reliable path from a source, i.e., the transmit sensor node, to a destination, i.e., the sink. To show the advantages of our proposed schemes, the scheduling scheme recently proposed in [35] is used as the benchmark scheme[2] for comparison purpose. It is noted that the benchmark scheme implemented in our considered system setup is more complicated than that in [35] which solely considered conventional multi-hop cognitive radio networks without energy harvesting and imperfect CSI of interference link. The main contributions of the paper can be summarized as follows:

- We consider a general multi-hop cognitive wireless powered D2D communication in WSNs with imperfection of interference channels, which is not available in the literature [32], [36], [37], [39]. We show that the imperfection of interference channels affects significantly any performance metric such as outage probability, outage floor, and performance loss of multi-hop cognitive D2D communication in WSNs.
- We propose DHS and BPS schemes to improve dual-hop diversity transmission and to minimize the end-to-end outage probability for considered system setup. In particular, the BPS scheme significantly outperforms DHS scheme, which in turns outperforms the benchmark scheme. The complexity analysis for each scheme is provided that allows us to design a proper scheduling strategy to achieve a good complexity-performance trade-off for specific D2D communication applications in WSNs such as search and rescue missions.
- We derive new closed-form expressions for the outage probability of each scheme under imperfection of interference channels. To gain more useful insights, we perform the performance floor analyses of all schemes for interference-limited and interference-tolerant scenarios. The performance floor analyses show that the imperfection of interference channels affects the interference-limited scenario but it does not affect the interference-tolerant one. The correctness of the developed analysis is then verified by Monte-Carlo simulation.

The rest of the paper is arranged as follows. Section II introduces the system model and the underlying fundamentals of the proposed schemes. Section III carries out the outage performance analysis of the considered scenarios. The performance floor analysis is provided in Section IV. Section V presents representative numerical results based on the derived analytical results. Finally, Section VI concludes the paper.

Mathematical Notations and Functions: $\Pr[.]$ symbolizes probability, $f_X(.)$ and $F_X(.)$ denote the probability density function (PDF) and cumulative distribution function (CDF) of random variable X, respectively. $\Gamma(.)$ represents the Gamma function [48, Eq. 8.310.1], and $K_v(.)$ stands for the v-order modified Bessel function of second kind [55, Eq. 9.6.22].

II. System Model

A. System Description

Let us consider multi-hop cognitive wireless powered D2D communications in WSNs, where the secondary multi-hop is allowed to operate cognitively on the licensed spectrum to the primary network, as depicted in Fig. 1. The primary network, which can be seen as cellular network, consists of primary transmitters (PTs)[3] and L primary receivers PR_l, $l \in \{1, \ldots, L\}$. In the secondary multi-hop WSN, a source (S), e.g., a certain sensor, transmits its data to a destination (D), e.g., a sink node, with the help of intermediate cooperative sensor nodes which are referred to as relays hereafter, located in $K-1$ clusters with $K > 1$. We assume that all sensor nodes are equipped with single antenna, operate in half-duplex mode, and support D2D communications. A relay, $\mathsf{R}_{k,n}$, is the nth relay node in the kth cluster belonging to a set of relays $\mathcal{R} = \{\mathsf{R}_{k,n} | k = 0, \ldots, K, n = 1, \ldots, N_k\}$. Note that $\mathsf{R}_{0,1} \equiv \mathsf{S}$ and $\mathsf{R}_{K,1} \equiv \mathsf{D}$. We assume that all sensor nodes have a limited power supply; hence, they have to harvest energy from RF signals radiated by a set of power beacons $\mathcal{P} = \{\mathsf{PB}_m | m = 1, \ldots, M\}$, located along the multi-hop secondary WSN. Consider a channel from X to Y, where $\mathsf{X} \in \{\mathsf{S}, \mathsf{PB}_m, \mathsf{R}_{k,n}\}$ and $\mathsf{Y} \in \{\mathsf{S}, \mathsf{R}_{k,n}, \mathsf{PR}_l, \mathsf{D}\}$, let u_{XY} be the channel coefficient of the $\mathsf{X} \rightarrow \mathsf{Y}$ link. We assume that all channels exhibit quasi-static independent identically distributed (i.i.d.) Rayleigh fading. Therefore, the channel gain $|u_{\mathsf{XY}}|^2$ follows an exponential distribution, whose CDF and PDF are expressed as $F_{|u_{\mathsf{XY}}|^2}(x) = 1 - \exp(-\frac{x}{\lambda_{\mathsf{XY}}})$ and $f_{|u_{\mathsf{XY}}|^2}(x) = \frac{1}{\lambda_{\mathsf{XY}}} \exp(-\frac{x}{\lambda_{\mathsf{XY}}})$, respectively, where λ_{XY} is the mean of $|u_{\mathsf{XY}}|^2$. To take into account the large-scale pathloss effect for D2D communications, λ_{XY} can be expressed as $\lambda_{\mathsf{XY}} = \mathcal{L}(d_{\mathsf{XY}}/d_0)^{-\beta}$ [33], [34], where d_{XY} is the distance (in meter) between X and Y, β denotes the pathloss exponent, d_0 is the reference distance and $\mathcal{L}$ is the measured pathloss at d_0. For the sake of notational convenience, we denote $\lambda_{E,k} = \lambda_{\mathsf{PB}_m \mathsf{R}_{k,i}}$, $\lambda_{I,k} = \lambda_{\mathsf{R}_{k,i} \mathsf{PR}_l}$, and $\lambda_{D,k} = \lambda_{\mathsf{R}_{k,i} \mathsf{R}_{k,j}}$ for energy harvesting, interference and data links, respectively.

[2]For convenience of presentation, we use the term "the benchmark scheme" instead of "the scheme in [35]" in some cases.

[3]We mainly focus on the cognitive multi-hop D2D communication in WSN and the impacts of multiple PRs on its performance; hence, considering the transmissions between PTs and PRs is beyond the scope of this paper.

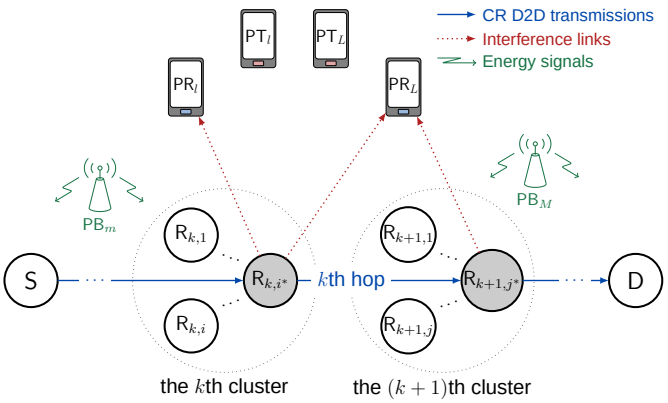

Fig. 1: Illustration of the proposed multi-hop cognitive wireless powered D2D communications in WSNs.

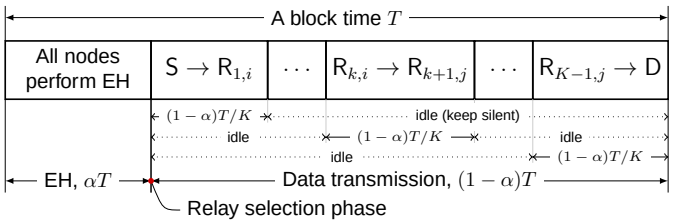

Fig. 2: The transmission block structure of the multi-hop cognitive wireless powered D2D communications in WSNs.

The system operation is divided into two consecutive phases, as sketched in Fig. 2. In phase 1, called the EH phase, all sensor nodes simultaneously harvest energy from multiple PBs based on the time-switching (TS) architecture [36] with duration of αT, where α and T denote the time switching ratio and the transmission block time[4], respectively. In phase 2, called the cooperative transmission phase, sensor nodes use cognitive D2D communication to transmit their data to the destination. In particular, phase 2 is separated into two sub-phases as follows. The first sub-phase conducts the user scheduling processes via control signal systems with the help of CSI and/or harvested energy information while the second sub-phase performs data transmission through K orthogonal sub-time slots. We assume that $\mathsf{R}_{k,i}$ and $\mathsf{R}_{k+1,j}$ are scheduled to participate at the kth hop. The relay, $\mathsf{R}_{k,i}$, decodes the signal received from the previous hop, and uses its harvested energy with cognitive radio constraint to forward the re-encoded signal to $\mathsf{R}_{k+1,j}$. The process is repeated at each hop, except that the 0th hop is $\mathsf{S} \to \mathsf{R}_{1,i}$ transmission and the $(K-1)$th hop is $\mathsf{R}_{K-1,j} \to \mathsf{D}$ transmission. The operation of TS architecture at each sensor $R_{k,n}$, selecting (scheduling) $\mathsf{R}_{k,i}$ and $\mathsf{R}_{k+1,j}$, and information processing at D, will be explained in detail next.

B. Energy Harvesting Phase and Interference Model

Considering the TS architecture, the energy harvested by $\mathsf{R}_{k,n}$ from M PBs can be expressed as $E_{k,n} = \sum_{m=1}^{M} \eta\alpha T P_m |g_{m,k,n}|^2$, where $\eta \in (0,1)$ is the energy conversion efficiency and $g_{m,k,i}$ is the EH channel coefficient of the link $\mathsf{PB}_m \to \mathsf{R}_{k,n}$. We assume that all harvested energy at each node stored in a super capacitor is fully consumed for information transmission [32], [36]. Thus, the transmit power of $\mathsf{R}_{k,n}$ based on energy harvested during a sub-time slot of $(1-\alpha)T/K$ can be expressed as

$$P_{k,n}^{\mathtt{EH}} = \sum_{m=1}^{M} \frac{K\eta\alpha}{(1-\alpha)} P_m |g_{m,k,n}|^2, \tag{1}$$

In underlay cognitive system, the interference power received at the primary user from the D2D transmitter must be lower than the peak interference power constraint, i.e., I_p, to protect the primary network operation. Since multiple PRs are deployed in the primary network, the maximum transmit power of relay node $\mathsf{R}_{k,n}$ can be determined as [53]

$$P_{k,n}^{\mathtt{I}} = I_p \big/ \max_{l=1,\ldots,L} |f_{k,n,l}|^2, \tag{2}$$

where $f_{k,n,l}$ denotes the interference channel coefficient of link $\mathsf{R}_{k,n} \to \mathsf{PR}_l$. In practice, the perfect knowledge of interference channel is difficult to be estimated accurately at the transmitter in secondary multi-hop WSNs due to complex channel intricacies such as the feedback delay transmission. Thus, $f_{k,n,l}$ can be estimated as $\hat{f}_{k,n,l}$, and mathematically expressed as [41] $\hat{f}_{k,n,l} = \rho f_{k,n,l} + \sqrt{1-\rho^2}\varepsilon_k$, where ε_k represents a circular symmetric complex Gaussian random variable with zero-mean and the same variance of $f_{k,n,l}$, and $\rho \in [0,1]$ is the correlation coefficient between the estimated channel and the real channel. The value of ρ is used to measure the CSI condition such as channel estimation error, mobility, limited feedback, and feedback delay [42]. The correlation coefficient can be determined using a pilot symbol modulated parameter (PSAM) [40]. Note that the higher value of ρ is, the lower channel estimation error is. The transmit power of the relay $\mathsf{R}_{k,i}$ with imperfect CSI can be expressed as

$$\widehat{P}_{k,n}^{\mathtt{I}} = I_p \big/ \max_{l=1,\ldots,L} |\hat{f}_{k,n,l}|^2. \tag{3}$$

From (1) and (3), the maximum allowable transmit power at $\mathsf{R}_{k,i}$ is constrained as

$$P_{k,n} = \min\left(\sum_{m=1}^{M} \kappa P_m |g_{m,k,n}|^2, I_p \big/ \max_{l=1,\ldots,L} |\hat{f}_{k,n,l}|^2 \right), \tag{4}$$

where $\kappa \triangleq K\eta\alpha/(1-\alpha)$. At the end of the EH process, the PBs turn to sleep mode during the data transmission time to save their energy and limit co-channel interference. Note that when relays can not consume all their harvested energies due to the constraint (4), the remaining energy will be leaked over time since each sensor node is equipped with an inexpensive capacitor and without the energy storage management [32].

C. Cooperative Transmission Phase

In this phase, we apply the time division multiple access scheduling for our considered system model to obtain the global synchronization and effective operation [22], [23]. Specifically, TDMA scheduling allows sensor nodes to switch into idle states until their allocated time slots, which avoids collisions, saves energy, and bounds the network delays [23]. In particular, the time duration for the data transmission is

[4]As within a time block T, the location of the sensor nodes and their connections do not change. Thus, the network topology is considered static but changes independently from one time block to another.

equally divided into K orthogonal sub-time slots and each sub-time slot is presented as $(1-\alpha)T/K$, as shown in Fig. 2. By using TDMA scheduling, each D2D node can access the frequency spectrum in the period of $(1-\alpha)T/K$ for its data transmission while other D2D relays switch into the idle mode. Therefore, the harvested energy is efficiently used and the WSN's lifetime is prolonged. In our system model, sensor nodes can collaborate and create virtual structures by grouping nearby neighbor nodes into clusters, where sensor nodes in a cluster share the same interests or behaviors [43]. The distributed clustering algorithms can be employed for the considered system allowing any sensor node to join a cluster depending on their locations, as discussed in [44], [45]. Before the data transmission phase, the relay selection process will be performed according to the MAC protocols [46]. The source node S first broadcasts the Request-To-Send message (RTS) to all relay nodes located in its radio range. Once the relay $\mathsf{R}_{k,i}$ in cluster k receives the RTS message, it becomes aware of the direct data transmission from S and estimate the instantaneous fading channel coefficients to the source. Then, $\mathsf{R}_{k,i}$ feeds back a Clear-To-Send (CTS) message which carries its direct transmission information to the source. Relying on the receipted message CTS with fading channel information, S can select the best cooperative relay in cluster k for data transmission in the next phase. Note that the best forwarder will be selected for D2D communications based on its channel quality from the transmitter. After setting up the connected D2D links in MAC layer, the dynamic source routing (DSR) protocol is used to establish a multi-hop D2D communication route from source to destination in network layer [22], [47]. After successfully receiving the data, D feeds back an Acknowledgment (ACK) message to S indicating the end of data transmission in a block time. If the connection is broken due to the mobility of nodes, the neighbor nodes can easily repair the data routing route by activating the route maintenance process in DSR protocol. In this study, we only focus on the physical layer transmission of the proposed user scheduling schemes with the assumption that routing protocol and clustering are fully supported by the upper layers. Consider the relay node $\mathsf{R}_{k,i}$ at hop k, with $i \in \{1,\ldots,N_k\}$, forwards the re-encoded signal to the relay, $\mathsf{R}_{k+1,j}$, with $j \in \{1,\ldots,N_{k+1}\}$. Thus, the received signal at $\mathsf{R}_{k+1,j}$ can be expressed as $y_{k,i,j} = \sqrt{P_{k,i}} h_{k,i,j} x_{k,i,j} + w_j$, where $h_{k,i,j}$ denotes the channel coefficients of link $\mathsf{R}_{k,i} \to \mathsf{R}_{k+1,j}$, $w_j \sim \mathcal{CN}(0,\sigma_j^2)$, herein $\mathcal{CN}(0,\sigma_j^2)$ is the circular symmetric complex Gaussian variable with zero-mean and variance σ_j^2. We assume that each node has the same noise power level, i.e., $\sigma_j^2 = \sigma^2$. And $x_{k,i,j}$ is the transmit signal from $\mathsf{R}_{k,i}$, wherein $\mathbb{E}[|x_{k,i,j}|^2] = 1$, with $\mathbb{E}[\cdot]$ denoting expectation. Note that the signal $x_{k,i,j}$ is decoded and forwarded by $\mathsf{R}_{k+1,j}$. From (4), the instantaneous SNR for hop $k+1$ can be expressed as $\gamma_k = P_{k,n}|h_{k,i,j}|^2$, where $\bar{\gamma}_P = P_m/\sigma^2$ and $\bar{\gamma}_I = I_p/\sigma^2$.

D. Relay Selection Criterion

1) Dual-Hop Scheduling Scheme: The dual-hop relay selection criterion aims to make robust the D2D transmission capability of two consecutive hops. The detail operation of feedback channels as well as the scheduling procedure for DHS scheme are presented in Fig. 3.

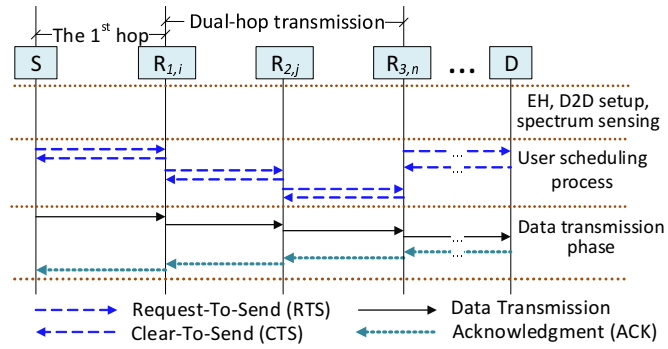

Fig. 3: User scheduling procedure for DHS scheme.

Consider two successive hops transmission including three clusters. Firstly, a relay in the third cluster providing the largest data channel gain to the transmitter among available ones will be the selected. Then, a relay in the second cluster will be chosen that maximizes the SNR between the first and the third selected relays. It is worth noting that the first selected relay has already been chosen in the previous dual-hop transmission. With this approach, two scenarios can be accounted: (i) the number of clusters is even; and (ii) the number of clusters is odd. In this paper, we consider the former scenario because it is a general scenario and includes the latter one. When the number of clusters is even, the number of hops is odd. Therefore, we consider separately the relay selection criteria for the first hop and the remaining $(K-1)/2$ hops. In the first hop transmission, a relay in cluster 1 which has the highest channel gain from the source will be selected. Thus, the relay in cluster 1 is chosen by the following criterion

$$\mathsf{R}_{1,j^*} = \arg \max_{j=1,\ldots,N_1} P_{0,1}|h_{0,1,j}|^2. \tag{5}$$

The instantaneous SNR in the first hop can be expressed as

$$\gamma_1 = \max_{j=1,\ldots,N_1} \min\left(\sum_{m=1}^{M} \kappa\bar{\gamma}_P|g_{m,0,1}|^2, \bar{\gamma}_I / \max_{l=1,\ldots,L} |\hat{f}_{0,1,l}|^2\right)|h_{0,1,j}|^2. \tag{6}$$

Next, consider the remaining $(K-1)/2$ hops. It is assumed that the selected relay at cluster $(2k-1)$ is R_{2k-1,u^*} with $k \in \{1,\ldots,(K-1)/2\}$, then the best relay at clusters $2k$ and $(2k+1)$ can be selected by the criterion (7), given at the top of the next page. By deploying the criterion in (7), the SNR of two consecutive hops is maximized and the dual-hop transmission is more reliable in multi-hop WSNs.

It is noteworthy that this analysis can be perfectly applied to the scenario in which the number of clusters is odd.

2) Best-Path Scheduling Scheme: In this scheme, a path relay providing the highest end-to-end SNR will be selected among available ones for data transmission. The detail operation of feedback channels as well as the scheduling procedure for BPS scheme can be presented in Fig. 4.

If N_i is the number of relays in cluster i, with $i \in \{1,\ldots,K-1\}$, the total number of available communication paths from S to D will be Ψ, where $\Psi \triangleq \prod_{i=1}^{K-1} N_i$. Therefore, the BPS

$$(\mathsf{R}_{2k,i^*}, \mathsf{R}_{2k+1,j^*}) = \arg \max_{i=1,\ldots,N_{2k}} \min\Big[\min\Big(\sum_{m=1}^{M} \kappa\bar{\gamma}_P |g_{m,2k-1,u^*}|^2, \bar{\gamma}_I / \max_{l=1,\ldots,L} |\hat{f}_{2k-1,u^*,l}|^2\Big)|h_{2k-1,u^*,i}|^2,$$
$$\max_{j=1,\ldots,N_{2k+1}} \min\Big(\sum_{m=1}^{M} \kappa\bar{\gamma}_P |g_{m,2k,i}|^2, \bar{\gamma}_I / \max_{l=1,\ldots,L} |\hat{f}_{2k,i,l}|^2\Big)|h_{2k,i,j}|^2\Big]. \quad (7)$$

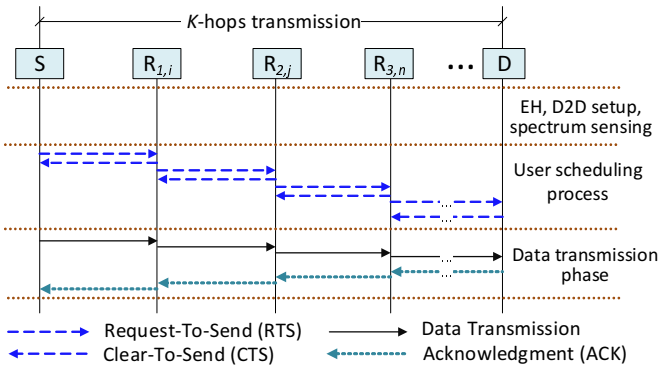

Fig. 4: User scheduling procedure for BPS scheme.

scheme tries to select the best path among Ψ available paths for data transmission. Mathematically, the selection criterion for the BPS scheme can be expressed as

$$(\mathsf{R}_{1,i^*}, \mathsf{R}_{2,i^*}, \ldots, \mathsf{R}_{K-1,i^*}) = \arg \max_{p=1,\ldots,\Psi} \min_{k=1,\ldots,K-1}$$
$$\Big[\min\Big(\sum_{m=1}^{M} \kappa\bar{\gamma}_P |g_{m,k,i}|^2, \bar{\gamma}_I / \max_{l=1,\ldots,L} |\hat{f}_{k,i,l}|^2\Big)|h_{k,i,j}|^2\Big], \quad (8)$$

where $p \in \{1, \ldots, \Psi\}$ is the pth communication path.

III. Outage Performance Analysis

In this section, the performance of the proposed system setup is analyzed and evaluated in terms of outage probability and outage floor. For the sake of notational convenience, we use some self-defined functions along the forthcoming analysis, and they are described follows:

$$\Xi(x,y,z) = \Big(\frac{x}{\kappa\bar{\gamma}_P\lambda_{E,k}y}\Big)^{\frac{z}{2}} K_z\Big(2\sqrt{\frac{x}{\kappa\bar{\gamma}_P\lambda_{E,k}y}}\Big),$$
$$\psi(x,y) = \binom{x}{y}(-1)^{y-1}, \chi(x,y,z) = \frac{xI_p}{(xI_p + yz)},$$

$$\Lambda(x,z) = \Big(\frac{x}{\kappa\bar{\gamma}_P\lambda_{D,k}\lambda_{E,k}} + \frac{\bar{\gamma}_I}{\kappa\bar{\gamma}_P\Omega_I\lambda_{E,k}}\Big)^{\frac{z}{2}}$$
$$\times K_z\Big(2\sqrt{\frac{x}{\kappa\bar{\gamma}_P\lambda_{D,k}\lambda_{E,k}} + \frac{\bar{\gamma}_I}{\kappa\bar{\gamma}_P\Omega_I\lambda_{E,k}}}\Big),$$
$$\varpi_k = \frac{\bar{\gamma}_I}{(\kappa\bar{\gamma}_P\Omega_{I,k}\lambda_{E,k})}, \vartheta(x,y) = \binom{x-1}{y}\frac{2x(-1)^y}{y+1}.$$

In the proposed multi-hop network with K-hop transmission, the information rate can be expressed as [41]

$$R_{\mathtt{sch}} = (1-\alpha)/K \log_2\big(1 + \min_{k=1,\ldots,K} \gamma_k\big). \quad (9)$$

where $(1-\alpha)/K$ indicates the information transmission period when the data transmission time is divided in K orthogonal time slots for multi-hop transmission. The OP of a certain communication can be defined as the probability that the information rate ($R_{\mathtt{sch}}$) falls below a predefined target data rate (R_{th}) [53]. Thus, the OP of proposed schemes can be expressed as

$$P_{\mathrm{out}}^{\mathtt{sch}} = \Pr\big[R_{\mathtt{sch}} < R_{\mathrm{th}}\big], \quad (10)$$

where $\mathtt{sch} \in \{\mathtt{DHS}, \mathtt{BPS}\}$.

A. Dual-Hop Scheduling Scheme

From the selection criteria as in (6) and (7), the OP of DHS scheme can be calculated by the following theorem.

Theorem 1: The exact closed-form expression for the OP of DHS scheme can be derived as

$$P_{\mathrm{out}}^{\mathtt{DHS}} = 1 - (1 - \mathcal{F}_{1,1} - \mathcal{F}_{2,1}) \prod_{k=1}^{(K-1)/2} \Big(1 - \prod_{i=1}^{N_{2k}} [1-$$
$$(1-\mathcal{F}_{3,2k-1} - \mathcal{F}_{4,2k-1})(1 - \mathcal{F}_{1,2k} - \mathcal{F}_{2,2k})]\Big), \quad (11)$$

where $\mathcal{F}_{1,1}$, $\mathcal{F}_{2,1}$, $\mathcal{F}_{3,2k-1}$, $\mathcal{F}_{4,2k-1}$, $\mathcal{F}_{1,2k}$ and $\mathcal{F}_{2,2k}$ can be expressed as (31), (33), (43), (44), (45) and (46), respectively.

Proof: The proof of Theorem 1 is in Appendix A. ■

B. Best-Path Scheduling Scheme

By using all CSI channels, the OP of BPS scheme can be analyzed by the following theorem.

Theorem 2: The exact closed-form expression for the OP of BPS scheme can be derived as

$$P_{\mathrm{out}}^{\mathtt{BPS}} = \prod_{p=1}^{\Psi} \Big[1 - \prod_{k=1}^{K-1} \Big(1 - \mathcal{F}_{1,k} - \mathcal{F}_{2,k}\Big)\Big], \quad (12)$$

where $\mathcal{F}_{1,k}$ and $\mathcal{F}_{2,k}$ are shown as (48) and (49), respectively.

Proof: The proof of Theorem 2 is in Appendix B. ■

In order to gain further insights into the system outage performance behaviors in different scenarios, the outage floor analysis of all schemes will be provided and discussed in the following section.

IV. Performance Floor Analysis

The outage floor of a certain cognitive communication system indicates that the outage probability can not fall below a given level even though the values of average SNRs are increased [49]. In this paper, we analyze the outage floor of all schemes to give insightful information into the system outage saturation phenomenon and the impact of imperfect interference channel to the multi-hop cognitive wireless powered D2D communications in WSNs. In particular, two typical deployment scenarios will possibly occur in practice. The first case corresponds to the interference-tolerant scenario, where

the PRs are located very far from multi-hop network, i.e., $I_p \gg P_m$. The second case is seen as the interference-limited scenario, where the position of primary users is nearby of the multi-hop network, i.e., $I_p \ll P_m$. Each situation causes some specific impacts on the network performance. From (4), the transmit power $P_{k,i}$ can be rewritten as

$$P_{k,i} = \min\Big(\sum_{m=1}^{M} \kappa P_m |g_{m,k,i}|^2, I_p / \max_{l=1,\ldots,L} |\hat{f}_{k,i,l}|^2\Big), \approx \begin{cases} \sum_{m=1}^{M} \kappa P_m |g_{m,k,i}|^2, \text{ if } & I_p \gg P_m, \\ I_p / \max_{l=1,\ldots,L} |\hat{f}_{k,i,l}|^2, \text{ if } & I_p \ll P_m. \end{cases} \tag{13}$$

A. Dual-Hop Scheduling Scheme

Theorem 3: The asymptotic expression for the outage floor of DHS scheme with interference-tolerant and interference-limited cases can be derived as

$$P_{\text{floor}}^{\texttt{DHS}} \approx \begin{cases} F_{\gamma_1^T}(\gamma_{\text{th}}) + \sum_{k=1}^{(K-1)/2} F_{\gamma_k^T}(\gamma_{\text{th}}), \text{ if } & I_p \gg P_m, \\ F_{\gamma_1^L}(\gamma_{\text{th}}) + \sum_{k=1}^{(K-1)/2} F_{\gamma_k^L}(\gamma_{\text{th}}), \text{ if } & I_p \ll P_m, \end{cases} \tag{14}$$

where $F_{\gamma_1^T}(\gamma_{\text{th}})$, $F_{\gamma_k^T}(\gamma_{\text{th}})$, $F_{\gamma_1^L}(\gamma_{\text{th}})$, and $F_{\gamma_k^L}(\gamma_{\text{th}})$ are given as (53), (54), (57), (58), respectively.

Proof: The proof of Theorem 3 is in Appendix C. ■

B. Best-Path Scheduling Scheme

Theorem 4: The asymptotic expression for the outage floor of BPS scheme with interference-tolerant and interference-limited cases can be derived as

$$P_{\text{floor}}^{\texttt{BPS}} \approx \begin{cases} \prod_{p=1}^{\Psi} \Big[\sum_{k=1}^{K-1} F_{\gamma_{1,k}^T}(\gamma_{\text{th}})\Big], \text{ if } & I_p \gg P_m, \\ \prod_{p=1}^{\Psi} \Big[\sum_{k=1}^{K-1} F_{\gamma_{1,k}^L}(\gamma_{\text{th}})\Big], \text{ if } & I_p \ll P_m, \end{cases} \tag{15}$$

where $F_{\gamma_{1,k}^T}(\gamma_{\text{th}})$ and $F_{\gamma_{1,k}^L}(\gamma_{\text{th}})$ are shown as (60) and (61), respectively.

Proof: The proof of Theorem 4 is in Appendix D ■

It is shown from (15) that when $I_p \gg P_m$, the $P_{\text{floor}}^{\texttt{BPS}}$ does not depend on the parameters of interference channels such as $\lambda_{I,k}$, L and $\bar{\gamma}_I$. Conversely, when $I_p \ll P_m$, $P_{\text{floor}}^{\texttt{BPS}}$ does not depend on the parameters of EH channels such as $\lambda_{E,k}$, M and $\bar{\gamma}_P$. The reason is that the transmit power $P_{k,i}$ is different in two cases, as shown in (13) resulting in the different outage floors and characteristics for BPS scheme.

V. Numerical Results

In this section, we present representative numerical results to illustrate the achievable performance of the proposed user scheduling schemes. We consider a line multi-hop secondary network in a square of unit area, where relays are equally coordinated between source S and destination D in the x-axis, as done in [37], [41]. For the sake of exposition, the positions of S, D, PB_m and PR_l are located, respectively, at $(0,0)$, $(0,30)$, $(15,5)$ and $(20,-5)$[5], which implies that the distance between two consecutive D2D relays is 10 meters [25], [26]. Unless otherwise stated, we set in all simulations the path-loss exponent $\beta = 2.7$, the EH efficiency $\eta = 0.8$, the time switching ratio $\alpha = 0.2$, the target data rate $R_{\text{th}} = 1$ (bps/Hz), the number of PRs $L = 3$, the number of PBs $M = 3$, number of hops $K = 3$, and the number of relays in each cluster $N_k = 2$.

To demonstrate the outstanding performance of proposed schemes, the relay selection scheme taken from [35], which is considered as the state-of-the-art solution, is used as a benchmark scheme for comparison purpose. We provide the exact closed-form expression for the outage probability of benchmark scheme that can be expressed as

$$P_{\text{out}}^{\texttt{BM}} = 1 - \Big[1 - \prod_{i=1}^{N_{K-1}} \Big(1 - (1 - \mathcal{F}_{3,K-2} - \mathcal{F}_{4,K-2}) \times (1 - \mathcal{F}_{3,K-1} - \mathcal{F}_{4,K-1})\Big)\Big] \prod_{k=1}^{K-2} \Big[1 - \mathcal{F}_{1,k} - \mathcal{F}_{2,k}\Big], \tag{16}$$

where $\mathcal{F}_{1,k}$, $\mathcal{F}_{2,k}$ can be obtained by simply replacing the index $2k$ as k in (45) and (46), respectively, and $\mathcal{F}_{3,A}$ and $\mathcal{F}_{4,A}$, with $A \in \{K-1, K-2\}$, can be obtained by replacing the index $2k-1$ as A in (43) and (44), respectively. We also provide the asymptotic expression for the outage floor of the benchmark scheme with interference-tolerant and interference-limited scenarios that can be expressed as

$$P_{\text{floor}}^{\texttt{BM}} \approx \begin{cases} F_{\gamma_{K-1}^T}(\gamma_{\text{th}}) + \sum_{k=1}^{K-2} F_{\gamma_k^T}(\gamma_{\text{th}}), \text{ if } & I_p \gg P_m, \\ F_{\gamma_{K-1}^L}(\gamma_{\text{th}}) + \sum_{k=1}^{K-2} F_{\gamma_k^L}(\gamma_{\text{th}}), \text{ if } & I_p \ll P_m, \end{cases} \tag{17}$$

where

$$F_{\gamma_k^T}(\gamma_{\text{th}}) = 1 - \sum_{n=1}^{N_k} \frac{2\psi(N_k, n)\Xi(n\gamma_{\text{th}}, \lambda_{D,k}, M)}{\Gamma(M)}, \tag{18}$$

$$F_{\gamma_{K-1}^T}(x) = \prod_{i=1}^{N_{K-1}} \Big[1 - \frac{4\Xi(\gamma_{\text{th}}, \lambda_{D,K-2}, M)}{\Xi(\gamma_{\text{th}}, \lambda_{D,K-1}, M)^{-1}\Gamma(M)^2}\Big], \tag{19}$$

$$F_{\gamma_k^L}(\gamma_{\text{th}}) = 1 - \sum_{l=0}^{L-1} \sum_{n=1}^{N_k} \frac{\vartheta(L,l)\psi(N_k,n)}{2\chi(\lambda_{D,k}, \gamma_{\text{th}}, \Omega_{I,k})^{-1}}, \tag{20}$$

$$F_{\gamma_{K-1}^L}(x) = \prod_{i=1}^{N_{K-1}} \Big[1 - \sum_{l=0}^{L-1} \vartheta(L,l)\chi(\lambda_{D,K-2}, \gamma_{\text{th}}, \Omega_{I,K-2}) \times \sum_{l=0}^{L-1} \vartheta(L,l)\chi(\lambda_{D,K-1}, \gamma_{\text{th}}, \Omega_{I,K-1})\Big]. \tag{21}$$

We would like to omit the derivation of expressions in (16) and (17) since it follows the similar manner as we have done in the analysis of our proposed schemes. It is also evident from expressions in (16) and (17) that the performance of benchmark scheme is presented in its new closed-form expressions, which have been not reported in the previous

[5]In our simulation, on one hand, the relays (beacons) are assumed to be located at the same place. It implies that these nodes have the same large-scale fading. On the other hand, channels related to the relays (beacons) are models as random variables which results in different corresponding small-scale fading. It is noteworthy that this assumption for the simulation has been widely adopted in the literature, as in [41], [49], [53].

works, since our proposed network model, which combines energy harvesting, underlay cognitive radio, and multi-relay multi-hop D2D communication, is unique and considered for the first time in the literature.

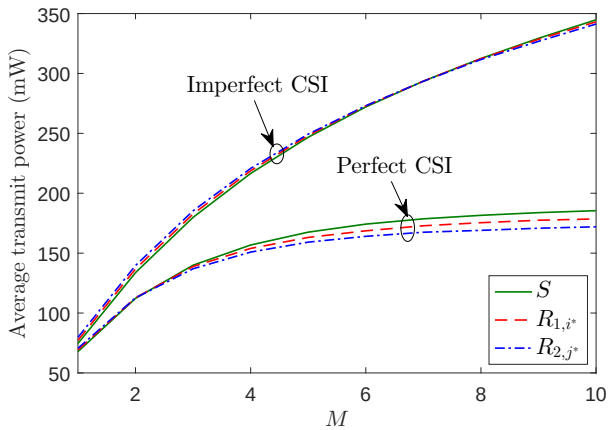

Fig. 5: Average transmit power versus M under different CSI conditions, with $\bar{\gamma}_I = 18$ dB and $\bar{\gamma}_P = 24$ dB

To comprehensively show the effect of imperfect CSI on the system performance, we consider the outage probabilities of all schemes under the imperfect CSI condition, i.e., $\rho = 0.6$, and perfect CSI condition, i.e., $\rho = 1$ in Figs. 5 to 11.

We first study the average transmit power in mW versus the number of PBs as shown in Fig. 5. As can be observed, the average transmit powers of each D2D node increases proportionally to the number of PBs. This implies that D2D nodes can harvest more energy when more PBs are deployed along the multi-hop WSNs. This results ensure that the D2D nodes are able to handle some services for D2D communication subject to the minimum power constraint in cognitive WSNs. Fig. 5 also shows that the transmit power with perfect CSI condition is lower than that of imperfect CSI. The reason is that when the estimated error is large, the transmit power $\widehat{P}^{\mathtt{I}}_{k,n}$ in (3) is also increased. Thus, the transmit power of D2D nodes in (4) is dominated by the power from EH $P^{\mathtt{EH}}_{k,n}$ in (1), leading to the high transmit power.

Fig. 6 depicts the system OP of all schemes versus $\bar{\gamma}_P$ in dB with perfect and imperfect CSI conditions. The value of $\bar{\gamma}_I$ is fixed while the average SNR $\bar{\gamma}_P$ is varied. There are some useful observations that can be drawn from Fig. 6. When the average SNR $\bar{\gamma}_P$ is increased, the outage probability of all schemes is enhanced and also receives a considerable influence from the CSI imperfection, particularly at high average SNR region. More specifically, in BPS scheme, the system OP under perfect CSI condition is similar to that of the imperfect CSI one when $\bar{\gamma}_P$ is less than $\bar{\gamma}_I$, indicating the insignificant effect of CSI imperfection. However, when $\bar{\gamma}_P$ is larger than $\bar{\gamma}_I$, the imperfect CSI remarkably impacts on system OP resulting in different outage floors of BPS scheme. We can observe a similar phenomenon in DHS and benchmark schemes. The reason is that when $\bar{\gamma}_P < \bar{\gamma}_I$, the system operates in interference-tolerant case, which implies that the system OP is independent of the interference channels

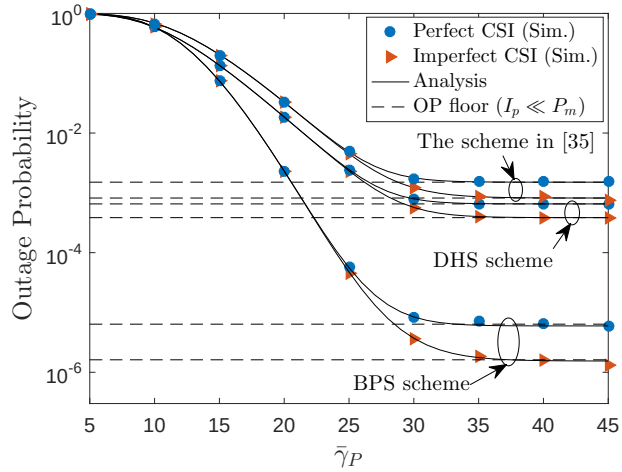

Fig. 6: The OP of three schemes versus $\bar{\gamma}_P$ in dB under different CSI conditions, with $\bar{\gamma}_I = 23$ dB

that was analytically expressed in Theorems 4, 5 and 6. Thus, the impact of imperfect CSI is negligible. Otherwise, when $\bar{\gamma}_P > \bar{\gamma}_I$, the system operates in interference-limited case, the system OP is dependent on the interference channels. Therefore, the impact of imperfect CSI is significant. Note that the outage floors are plotted in Fig. 6 based on the interference-limited case, i.e., $I_p \ll P_m$. The outage floors occur in all schemes because the transmit power of each D2D transmitter is always constrained as (4).

In underlay cognitive radio network, when the interference channel is estimated inaccurately, the actual transmit power in (3) dominates over the maximum allowable transmit power of D2D relays in (2), resulting in lower outage floor of all schemes for the imperfect CSI case. However, this is an undesirable situation from the primary network perspective because the exceeded transmit power causes interference to the PRs which often needs to be protected. The solution for such a problem will be presented in detail in Fig. 11. In protocol design perspective, the BPS scheme provides the best outage performance among that of all proposed schemes. The reason is that the more relays are jointly involved in the user scheduling process, the better performance the system has. Particularly, in BPS scheme, all relays are jointly taken part into the user scheduling process, while in the DHS scheme only relays in two consecutive clusters are participated in each user scheduling process. Thus, the BPS scheme clearly outperforms the DHS one. In the scheme in [35], only relays in a cluster participating in each scheduling process results in the worst performer. It is worth mentioning that the BPS scheme requires more CSI estimations than other ones leading the increase of system overhead. However, the BPS provides the best data transmission reliability among that of all schemes. The advantages and drawbacks of each scheme will be further discussed in the sequence of the simulation results.

Fig. 7 shows the system outage probability as a function of $\bar{\gamma}_I$ in dB. We vary the value of $\bar{\gamma}_I$ while the average SNR $\bar{\gamma}_P$ is fixed. As can be seen, the imperfect CSI only affects the system OP in moderate value of $\bar{\gamma}_I$. In particular, when $\bar{\gamma}_I$

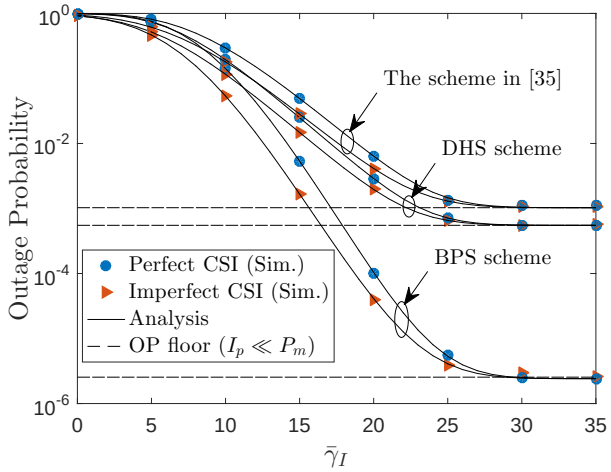

Fig. 7: The system outage probability versus $\bar{\gamma}_I$ in dB under different CSI conditions, with $\bar{\gamma}_P = 28$ dB

is less than $\bar{\gamma}_P$, the OP of all schemes in the imperfect CSI case is lower than that of perfect CSI case. However, when $\bar{\gamma}_I$ is larger than $\bar{\gamma}_P$, the OP of each scheme converges to its outage floor with negligible effect of imperfect CSI. Logically, this observation is consistent with what was shown in Fig. 6. Note that the outage floors is plotted in this figure based on the interference-tolerant case, i.e., $I_p \gg P_m$. Again, the BPS scheme shows its superiority over other existing schemes. At high value of $\bar{\gamma}_I$, all schemes converge to their outage floors, which correctly validates the performance floor analysis approaches. Besides, Figs. 6 and 7 also show that the theoretical results are in excellent agreement with the simulation ones, validating the developed framework and analysis.

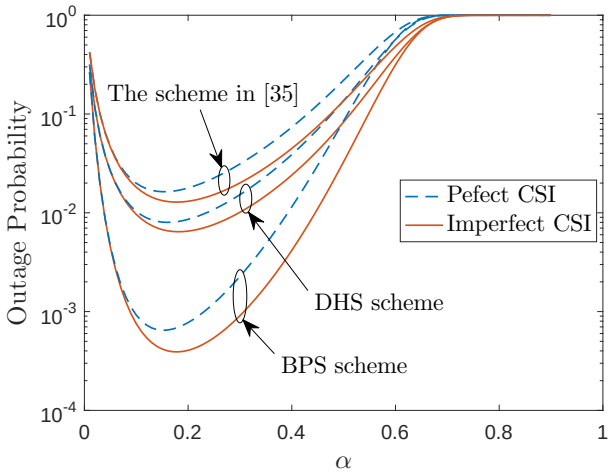

Fig. 8: The system OP versus α under different CSI conditions, with $\bar{\gamma}_I = 18$ dB and $\bar{\gamma}_P = 24$ dB

Fig. 8 presents the system OP as a function of α. As can be observed, the system OP is a convex function of α; hence, there exists an optimal value of α that minimizes the OP. More importantly, the imperfect CSI has no any effect on the system OP when α is relatively low. But it considerably impacts on the OP of all schemes when α is larger than about 0.05 resulting in different optimal value of α. The reason is that the transmit power of D2D nodes is governed by the amount of energy harvested from PBs in (4) when α is small. Otherwise, when α is larger than about 0.05, the ratio of peak interference power I_p to the interference channel gain dominates in (4) leading to the impact of imperfect CSI on system performance. Moreover, the system OP of all schemes seems to be saturated at the value of α which is greater than 0.7 in both perfect and imperfect CSI cases. Thus, appropriately designing value of α plays a vital role in the network planning to improve the overall system performance. Again, the BPS scheme is more robust than other existing ones, which shows the advantage of best-path relay selection criterion.

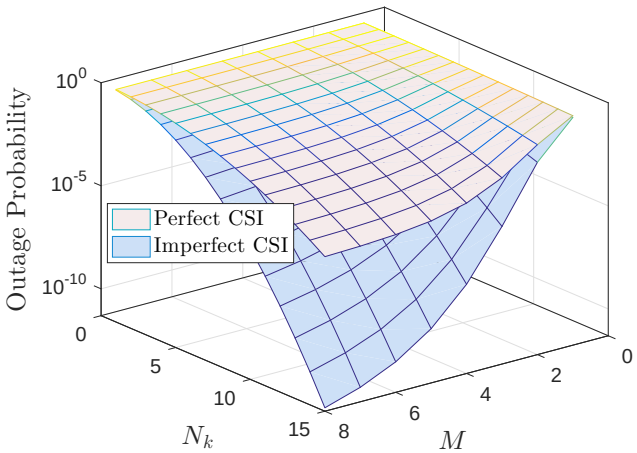

Fig. 9: Effect of N_k and M on the OP of BPS scheme under different CSI conditions, with $\bar{\gamma}_I = 4$ dB and $\bar{\gamma}_P = 10$ dB

The effects of number of power beacons, M, and the number of relays in each cluster, N_k, on the outage probability of BPS scheme are presented in Fig. 9. As can be observed, the enhancement of system OP is proportional to the number of PBs or number of relays under either perfect or imperfect CSI conditions. The reason is that when the number of relays in each cluster is increased, the end-to-end diversity transmission is improved, which in turn reduces the system outage probability. In addition, more power beacons are deployed, the amount of energy harvested at each sensor node is increased leading to the improvement of system performance. It is worth mentioning that the outage probability in the imperfect CSI case is always lower than that of perfect CSI one, which was clarified in Figs. 6 and 7.

Fig. 10 examines the effect of number of primary users, L, on the secondary network performance in the imperfect CSI condition. As can be observed, if only one PR is deployed in primary network, i.e., $L = 1$, there is no impact of imperfect CSI estimation on the multi-hop network performance. However, if multiple PRs are deployed, the performance of multi-hop network is degraded and highly depend on the imperfect CSI conditions. In particular, when the system has high accurate CSI estimation, i.e., $\rho = 1$, the OPs of all schemes are increased proportionally to the number of PRs. However,

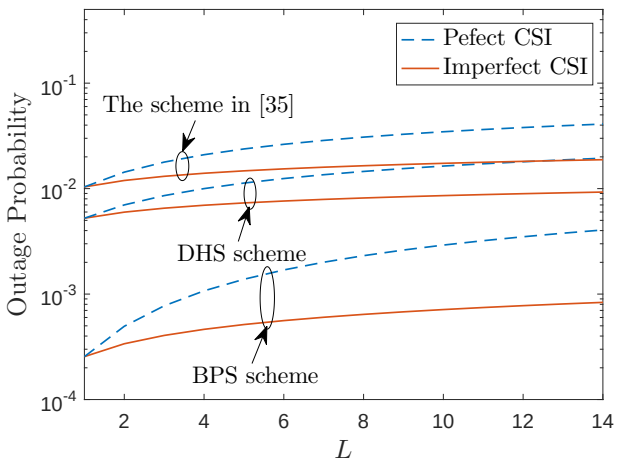

Fig. 10: The OP of three schemes versus L under different CSI conditions, with $\bar{\gamma}_I = 18$ dB and $\bar{\gamma}_P = 24$ dB

when the system undergoes low accurate CSI estimation, i.e., $\rho = 0.6$, the OP of all schemes seems to be slightly changed. For example, when L is varied from 1 to 14, the system OP of BPS scheme is increased from 2.6×10^{-4} to 4×10^{-3} with $\rho = 1$, and it is increased from 2.6×10^{-4} to 8×10^{-4} with $\rho = 0.6$. The reason is that the system still maintains the high transmit power in (4) when the estimated error is large, leading to the small change in the outage performance of multi-hop network. This is an undesired situation from the primary users perspective because the D2D nodes consume much unnecessary energy and induce severe interference to the PRs. Thus, the effects of this undesired behavior will be quantified and explained in the next figure.

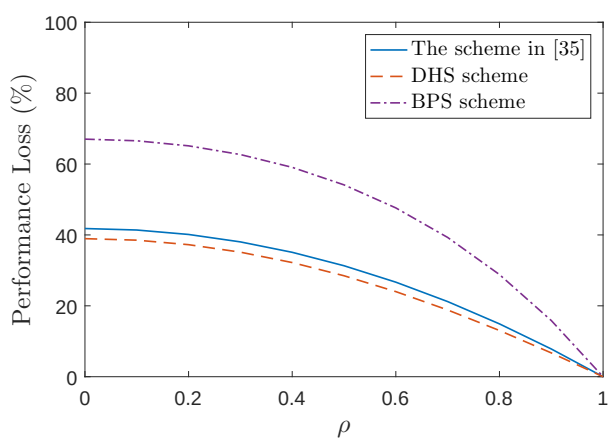

Fig. 11: The outage performance loss (%) of three schemes versus ρ with $\bar{\gamma}_I = 18$ dB and $\bar{\gamma}_P = 24$ dB

In underlay cognitive network, when the interference channel is estimated inaccurately, the actual interference caused to the primary users exceeds the level of I_p. These are undesirable effects from the primary network point of view and a possible solution is to reduce the transmit power constraint in (4) from the multi-hop network. Fig. 11 shows the outage performance loss as a function of ρ due to such a solution. The performance loss for the secondary network is defined as [42]

$$P_{\text{out}}^{\text{Loss}} = \frac{(P_{\text{out}}^{\text{Desired}} - P_{\text{out}}^{\text{Imperfect}})}{P_{\text{out}}^{\text{Desired}}}, \tag{22}$$

where $P_{\text{out}}^{\text{Desired}}$ and $P_{\text{out}}^{\text{Imperfect}}$ are the outage performance at secondary network for the perfect and imperfect CSI, respectively. As can be observed, the performance loss of all scheme is considerably increased when the estimate error of the interference channel is high, i.e., for a small value of ρ. Besides, when the estimate error is very small, i.e., $\rho = 1$, the performance loss of all scheme is reduced to zero. For example, at the largest estimate error, i.e., $\rho = 0$, the BPS, DHS and the benchmark schemes show the performance loss roughly 67%, 39%, and 42%, respectively. Surprisingly, the performance loss of benchmark scheme is higher than that of DHS one proving the outstanding performance of the proposed DHS scheme over the state-of-the-art solution.

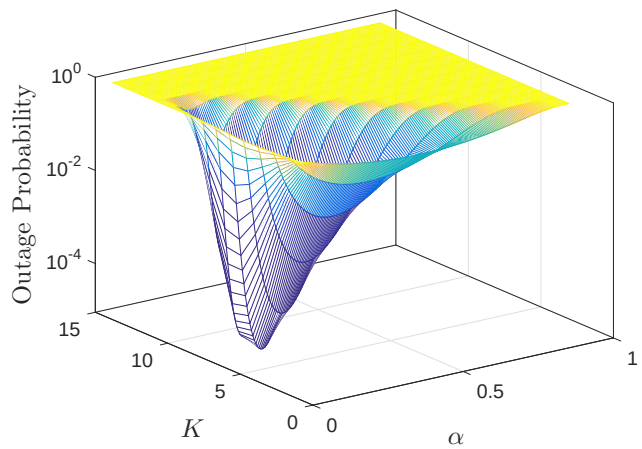

Fig. 12: Effect of K and α on the OP of BPS scheme with ρ = 0.6, $\bar{\gamma}_I = 14$ dB, and $\bar{\gamma}_P = 18$ dB

Fig. 12 shows the effect of the number of hops, K and the time switching ratio, α, on the system outage performance of BPS scheme. As can be observed, the system OP has the shape of convex and unimodal surface presenting a relation between K and α. In particular, when K is increased, the error probability for transmitting information to destination increases resulting in low system performance. Moreover, when α is large, the system requires much more time for data transmission, or when α is too small, the system does not have enough time for EH. These scenarios lead to the reduction of EH ability and degradation of system performance. Thus, K and α affect significantly on outage performance and play an important role in the design and network planing. Furthermore, we propose a numerical method to obtain optimal values of K and α to minimize the system OP. Adopting the golden section search [50] with minimum interval $\epsilon = 10^{-5}$, the optimal values of K and α are obtained as 6 and 0.1204, respectively, at the minimum value of OP as 1.55×10^{-5}. It is noted that the proposed numerical method can be feasibly applied for other schemes such as the benchmark and DHS.

TABLE I: The complexity order of all schemes

Type \ Scheme	Benchmark	DHS	BPS
The conventional network model in [51]	$\mathcal{O}(N_{\mathtt{con}}^{\mathrm{BM}})$	$\mathcal{O}(N_{\mathtt{con}}^{\mathrm{DHS}})$	$\mathcal{O}(N_{\mathtt{con}}^{\mathrm{BPS}})$
The wireless powered multi-hop network model [52]	$\mathcal{O}(N_{\mathtt{WPT}}^{\mathrm{BM}})$	$\mathcal{O}(N_{\mathtt{WPT}}^{\mathrm{DHS}})$	$\mathcal{O}(N_{\mathtt{WPT}}^{\mathrm{BPS}})$
The proposed multi-hop network model	$\mathcal{O}(N_{\mathtt{pro}}^{\mathrm{BM}})$	$\mathcal{O}(N_{\mathtt{pro}}^{\mathrm{DHS}})$	$\mathcal{O}(N_{\mathtt{pro}}^{\mathrm{BPS}})$

Finally, we provide the complexity order of the proposed schemes with different network types, as shown in Table I. The complexity order is defined by the number of channels that requires estimation. For comparison purposes, we also present two additional network types which are conventional multi-hop network without energy harvesting [51] and the wireless powered multi-hop network without cognitive radio [52]. We assume that each cluster has the same number of relays in two network types. The complexity orders of considered schemes, i.e., the benchmark, DHS and BPS schemes, for three network types are provided in Table I, where the values shown in the column Benchmark are given as $N_{\mathtt{con}}^{\mathrm{BM}} = (K-2)N_k + 2N_k$, $N_{\mathtt{WPT}}^{\mathrm{BM}} = (K-2)N + KM + 2N_k$, $N_{\mathtt{pro}}^{\mathrm{BM}} = (K-2)N_k + K(M+L) + 2N_k$, the values presented in the column DHS are given as $N_{\mathtt{con}}^{\mathrm{DHS}} = 3(K-1)N_k/2$, $N_{\mathtt{WPT}}^{\mathrm{DHS}} = 3N_k(K-1)/2 + KM$, $N_{\mathtt{pro}}^{\mathrm{DHS}} = 3N_k(K-1)/2 + K(M+L)$, and the values presented in the column BPS are given as $N_{\mathtt{con}}^{\mathrm{BPS}} = (K-2)N_k^2 + 2N_k$, $N_{\mathtt{WPT}}^{\mathrm{BPS}} = (K-2)N_k^2 + 2N_k + KM$, $N_{\mathtt{pro}}^{\mathrm{BPS}} = (K-2)N_k^2 + 2N_k + K(M+L)$. As can be observed from the column BPS in Table I, the number of CSI estimations required for the proposed network model, i.e., $N_{\mathtt{pro}}^{\mathrm{BPS}}$, is always higher than that of other network types. The reason is that our network model is general model considering both WPT and cognitive radio techniques, thus it is more complex than those of the network models in [51] and [52]. Furthermore, the required CSI for the BPS scheme is much larger than that of DHS and benchmark schemes in all network types. For example, in large scale network, $K = 10$, $N_k = 10$, $L = 10$ and $M = 10$, the amount of CSI channels for BPS, DHS and benchmark schemes in the proposed network are 1020, 335, and 300, respectively. In the network design perspective, each scheme, i.e., DHS and BPS schemes, should be judiciously applied on practical networks suitable with their hardware processing ability in order to achieve a good trade-off between performance gain and operation complexity.

VI. Conclusions

In this paper, we proposed user scheduling schemes to improve the outage performance for multi-hop cognitive wireless powered transfer D2D communications in WSNs. Particularly, we derived the exact closed-form expressions for the OP of benchmark, DHS and BPS schemes. The outage performance results were also analyzed under two limiting cases such as interference-tolerant and interference-limited, which are shown in simplified forms suitable for network planing and designing. The numerical results showed that for the same channel setting, the BPS scheme provided the best performance. Moreover, we showed that the inaccurate CSI estimation of interference links affects significantly any aspect of multi-hop D2D network in terms of outage probability, outage floor, and performance loss. In addition, the optimal pair of K and α could be obtained by using numerical results. As a result, the proposed schemes are indeed promising for future wireless sensor networks to enhance the system performance and extend the network coverage.

Appendix A
Proof of Theorem 1

From (10), the OP of the DHS scheme can be rewritten as

$$P_{\mathrm{out}}^{\mathtt{DHS}} = 1 - \left[1 - F_{\gamma_1}(\gamma_{\mathrm{th}})\right] \prod_{k=1}^{(K-1)/2} \left[1 - F_{\gamma_k}(\gamma_{\mathrm{th}})\right]. \quad (23)$$

For the sake of simplicity, let $X_1 = \sum_{m=1}^{M} |g_{m,0,1}|^2$, $Y_{1,j^*} = \max_{j=1,\ldots,N_k} |h_{0,1,j}|^2$ and $Z_{1,l^*} = \max_{l=1,\ldots,L} |\hat{f}_{0,1,l}|^2$. From (6), $F_{\gamma_1}(\gamma_{\mathrm{th}})$ can be calculated as

$$\begin{aligned} F_{\gamma_1}(\gamma_{\mathrm{th}}) =& \Pr[\kappa\bar{\gamma}_P X_1 < \bar{\gamma}_I / Z_{1,l^*}, \kappa\bar{\gamma}_P X_1 Y_{1,j^*} < \gamma_{\mathrm{th}}] \\ &+ \Pr[\kappa\bar{\gamma}_P X_1 > \bar{\gamma}_I / Z_{1,l^*}, \bar{\gamma}_I Y_{1,j^*} / Z_{1,l^*} < \gamma_{\mathrm{th}}]. \quad (24) \end{aligned}$$

The first term in the right hand side (RHS) of (24) can be calculated as

$$\mathcal{F}_{1,1} = \int_0^\infty F_{Z_{1,l^*}}\Big(\frac{\bar{\gamma}_I}{\kappa\bar{\gamma}_P x}\Big) F_{Y_{1,j^*}}\Big(\frac{\gamma_{\mathrm{th}}}{\kappa\bar{\gamma}_P x}\Big) f_{X_1}(x) dx. \quad (25)$$

Recalling that X_1 is the summation of M independent exponential random variables (RVs), thus it follows the chi-square distribution. Hence, the PDF and CDF of X_1 are given, respectively, as [53, Eq. (11)]

$$f_{X_1}(x) = \Big(\frac{1}{\lambda_{E,1}}\Big)^M \frac{x^{M-1}}{\Gamma(M)} \exp\Big(-\frac{x}{\lambda_{E,1}}\Big), \quad (26)$$

$$F_{X_1}(x) = 1 - \exp\Big(-\frac{x}{\lambda_{E,1}}\Big) \sum_{t=0}^{M-1} \frac{1}{t!} \Big(\frac{x}{\lambda_{E,1}}\Big)^t. \quad (27)$$

Since Y_{1,j^*} is the maximum of N_1 independent exponential RVs, the CDF of Y_{1,j^*} can be expressed as

$$\begin{aligned} F_{Y_{1,j^*}}(x) &= \sum_{n=0}^{N_1} \binom{N_1}{n} (-1)^n \exp\Big(-\frac{nx}{\lambda_{D,1}}\Big) \\ &= 1 - \sum_{n=1}^{N_1} \binom{N_1}{n} (-1)^{n-1} \exp\Big(-\frac{nx}{\lambda_{D,1}}\Big). \quad (28) \end{aligned}$$

Furthermore, the PDF of Y_{1,j^*} can be attained as $f_{Y_{1,j^*}}(x) = \sum_{n=1}^{N_1} (-1)^{n-1} \binom{N_1}{n} \frac{n}{\lambda_{D,1}} \exp(-\frac{nx}{\lambda_{D,1}})$. Next, the PDF and CDF of Z_{1,l^*} can be given, respectively, as [54, Eqs. (28) and (30)]

$$f_{Z_{1,l^*}}(x)=\sum_{l=0}^{L-1}\frac{\vartheta(L,l)}{2\Omega_{I,1}}\exp\Big(-\frac{x}{\Omega_{I,1}}\Big), \tag{29}$$

$$F_{Z_{1,l^*}}(x)=1-\sum_{l=0}^{L-1}\frac{\vartheta(L,l)}{2}\exp\Big(-\frac{x}{\Omega_{I,1}}\Big). \tag{30}$$

Substituting $f_{X_1}(\cdot)$, $F_{Z_{1,l^*}}(\cdot)$ and $F_{Y_{1,j^*}}(\cdot)$ into (25), along with the help of [48, Eq. (3.471.9)] and after some manipulations, $\mathcal{F}_{1,1}$ can be obtained as

$$\mathcal{F}_{1,1}=1-\frac{1}{\Gamma(M)}\Big[\sum_{n=1}^{N_1}\frac{2\psi(N_1,n)}{\Xi(n\gamma_{\rm th},\lambda_{D,1},M)^{-1}}+\sum_{l=0}^{L-1}\vartheta(L,l)\times\Xi(\bar{\gamma}_I,\Omega_{I,1},M)-\sum_{l=0}^{L-1}\sum_{n=1}^{N_1}\frac{\vartheta(L,l)\psi(N_1,n)}{\Lambda(n\gamma_{\rm th},M)^{-1}}\Big]. \tag{31}$$

The second term in the RHS of (24) can be expressed as

$$\mathcal{F}_{2,1}=\int_0^\infty\Big[1-F_{X_1}\Big(\frac{\bar{\gamma}_I}{\kappa\bar{\gamma}_P x}\Big)\Big]F_{Y_{1,j^*}}\Big(\frac{\gamma_{\rm th}x}{\bar{\gamma}_I}\Big)f_{Z_{1,l^*}}(x)dx. \tag{32}$$

Plugging $F_{X_1}(\cdot)$, $f_{Z_{1,l^*}}(\cdot)$ and $F_{Y_{1,j^*}}(\cdot)$ into (32), along with the help of [48, Eq. (3.471.9)] and after some manipulations, $\mathcal{F}_{2,1}$ can be obtained as

$$\mathcal{F}_{2,1}=-\sum_{t=0}^{M-1}\sum_{n=0}^{N_1}\sum_{l=0}^{L-1}\frac{\varpi_1\vartheta(L,l)\psi(N_1,n)}{t!\Lambda(n\gamma_{\rm th},t-1)^{-1}}. \tag{33}$$

For the CDF of γ_k in (23), let $X_{2k}=\sum_{m=1}^{M}|g_{m,2k,i^*}|^2$, $Y_{2k,j}=|h_{2k,i,j}|^2$ and $Z_{2k}=\max\limits_{l=1,\ldots,L}|\hat{f}_{2k,i^*,l}|^2$, $F_{\gamma_k}(\gamma_{\rm th})$ can be calculated by the following probability

$$F_{\gamma_k}(x)=\sum_{u=1}^{N_{2k-1}}\underbrace{\Pr(\mathsf{R}_{2k-1,u}=\mathsf{R}_{2k-1,u^*})}_{\Theta_{2k-1}}\Delta, \tag{34}$$

where Δ can be expressed as

$$\Delta=\prod_{i=1}^{N_{2k}}\Pr\Big[\min\Big(\min\Big(\kappa\bar{\gamma}_P X_{2k-1}Y_{2k-1,i},\frac{\bar{\gamma}_I Y_{2k-1,i}}{Z_{2k-1}}\Big),\max_{j=1,\ldots,N_{2k+1}}\min\Big(\kappa\bar{\gamma}_P X_{2k}Y_{2k,j},\bar{\gamma}_I Y_{2k,j}/Z_{2k}\Big)\Big)\Big]. \tag{35}$$

In this scheme, the best relay R_{2k-1,i^*} is selected by the strategy (5) to forward the re-encoded signal to the relays in cluster $2k$; thus, Θ_{2k-1} can be calculated as

$$\Theta_{2k-1}=\Pr\Big[\bigcap_{j=1,j\neq i}^{N_{2k-1}}(Y_{2k-1,i}>Y_{2k-1,j})\Big]=\int_0^\infty\prod_{j=1,j\neq i}^{N_{2k-1}}F_{Y_{2k-1,j}}(x)f_{Y_{2k-1,i}}(x)dx. \tag{36}$$

Plugging the CDF of $Y_{2k-1,j}$ and the PDF of $Y_{2k-1,i}$ into (36), along with the help of Binomial theorem, Θ_{2k-1} can be obtained as

$$\Theta_{2k-1}=\sum_{n=0}^{N_{2k-1}-1}\binom{N_{2k-1}-1}{n}\frac{(-1)^n}{n+1}\overset{(a)}{=}\frac{1}{N_{2k-1}}, \tag{37}$$

where (a) is due to [48, Eq. (0.155.1)]. Next, Δ can be calculated as (38), shown at the top of the next page. From (38), we also have $\Theta_{2k+1}=1/N_{2k+1}$. Based on the relay selection strategy in (7), ξ_{2k} can be derived as

$$\xi_{2k}=\Pr\Big[\bigcap_{l=1,l\neq i}^{N_{2k}}\Big(\min(Y_{2k-1,i},Y_{2k,j})>\min(Y_{2k-1,l},Y_{2k,j})\Big)\Big]\overset{(b)}{=}\int_0^\infty\prod_{l=1,l\neq i}^{N_{2k}}\Big(\Pr\Big[\min(Y_{2k-1,l},Y_{2k,j})<x\Big]\Big)f_U(x)dx, \tag{39}$$

where (b) is obtained by conditioning on $U=\min(Y_{2k-1,i},Y_{2k,j})$. The CDF of U can be obtained after some manipulations as

$$F_U(x)=\Pr\Big[\min(Y_{2k-1,i},Y_{2k,j})<x\Big]=1-\exp(-\lambda_U x), \tag{40}$$

and its PDF is obtained as

$$f_U(x)=\lambda_U\exp(-\lambda_U x). \tag{41}$$

where $\lambda_U\triangleq 1/\lambda_{D,2k-1}+1/\lambda_{D,2k}$. By plugging $f_U(x)$ into (39), along with the help of Binomial theorem, we arrive at

$$\xi_{2k}=\sum_{n=0}^{N_{2k}-1}\binom{N_{2k}-1}{n}(-1)^n\int_0^\infty\lambda_U\exp(-\lambda_U(n+1)x)=1/N_{2k}. \tag{42}$$

Similar to the approach developed for $F_{\gamma_1}(\gamma_{\rm th})$ in (24), I_1 and I_2 can be obtained, respectively, as $I_1=\mathcal{F}_{3,2k-1}+\mathcal{F}_{4,2k-1}$, and $I_2=\mathcal{F}_{1,2k}+\mathcal{F}_{2,2k}$, where

$$\mathcal{F}_{3,2k-1}=1-\frac{1}{\Gamma(M)}\Big[2\Xi(\gamma_{\rm th},\lambda_{D,2k-1},M)+\sum_{l=0}^{L-1}\vartheta(L,l)\times\Xi(\bar{\gamma}_I,\Omega_{I,2k-1},M)-\sum_{l=0}^{L-1}\vartheta(L,l)\Lambda(\gamma_{\rm th},M)\Big], \tag{43}$$

$$\mathcal{F}_{4,2k-1}=\sum_{t=0}^{M-1}\sum_{l=0}^{L-1}\frac{[\Xi(\bar{\gamma}_I,\Omega_{I,2k-1},t-1)-\Lambda(\gamma_{\rm th},t-1)]}{t!\varpi_{2k-1}^{-1}\vartheta(L,l)^{-1}}, \tag{44}$$

$$\mathcal{F}_{1,2k}=1-\frac{1}{\Gamma(M)}\Big[\sum_{n=1}^{N_{2k}}\frac{2\psi(N_{2k},n)}{\Xi(n\gamma_{\rm th},\lambda_{D,2k},M)^{-1}}+\sum_{l=0}^{L-1}\vartheta(L,l)\times\Xi(\bar{\gamma}_I,\Omega_{I,2k},M)-\sum_{l=0}^{L-1}\sum_{n=1}^{N_{2k}}\frac{\vartheta(L,l)\psi(N_{2k},n)}{\Lambda(n\gamma_{\rm th},M)^{-1}}\Big], \tag{45}$$

$$\mathcal{F}_{2,2k}=-\sum_{t=0}^{M-1}\sum_{n=0}^{N_{2k}}\sum_{l=0}^{L-1}\frac{\varpi_{2k}\vartheta(L,l)\psi(N_{2k},n)}{t!\Lambda(n\gamma_{\rm th},t-1)^{-1}}. \tag{46}$$

Now, plugging I_3 and I_4 into (38), where Δ is obtained along with $F_{\gamma_k}(x)$. Finally, substituting $F_{\gamma_1}(x)$ and $F_{\gamma_k}(x)$ into (23), we obtain the desired result as (11), which completes the proof of Theorem 1.

APPENDIX B
PROOF OF THEOREM 2

From (10), the OP of the BPS scheme can be rewritten as

$$P_{\rm out}^{\rm BPS}=\prod_{p=1}^{\Psi}\Big[1-\prod_{k=1}^{K-1}\big(1-F_{\gamma_k}(\gamma_{\rm th})\big)\Big], \tag{47}$$

Similar to the approach developed for I_1 in Theorem 1, $F_{\gamma_k}(\gamma_{\rm th})$ can be obtained as $F_{\gamma_k}(\gamma_{\rm th})=\mathcal{F}_{3,k}+\mathcal{F}_{4,k}$, where

$$\Delta = \sum_{j=1}^{N_{2k+1}} \underbrace{\Pr(\mathsf{R}_{2k+1,j} = \mathsf{R}_{2k+1,j^*})}_{\Theta_{2k+1}} \prod_{i=1}^{N_{2k}} \Bigg[1 - \Bigg(1 - \underbrace{\Pr\left[\min\left(\kappa\bar{\gamma}_P X_{2k-1} Y_{2k-1,i}, \frac{\bar{\gamma}_I Y_{2k-1,i}}{Z_{2k-1}}\right) < x\right]}_{I_1}\Bigg)$$
$$\times \Bigg(1 - \sum_{i=1}^{N_{2k}} \underbrace{\Pr(\mathsf{R}_{2k,i} = \mathsf{R}_{2k,i^*})}_{\xi_{2k}} \underbrace{\Pr\left[\max_{j=1,\ldots,N_{2k+1}} \min\left(\kappa\bar{\gamma}_P X_{2k} Y_{2k,j}, \frac{\bar{\gamma}_I Y_{2k,j}}{Z_{2k}}\right) < x\right]}_{I_2}\Bigg)\Bigg]. \quad (38)$$

$$\mathcal{F}_{3,k} = 1 - \frac{1}{\Gamma(M)}\Big[2\Xi(\gamma_{\rm th}, \lambda_{D,k}, M) + \sum_{l=0}^{L-1} \vartheta(L,l) \times \Xi(\bar{\gamma}_I, \Omega_{I,k}, M) - \sum_{l=0}^{L-1} \vartheta(L,l)\Lambda(\gamma_{\rm th}, M)\Big], \quad (48)$$

$$\mathcal{F}_{4,k} = \sum_{t=0}^{M-1}\sum_{l=0}^{L-1} \frac{\left[\Xi(\bar{\gamma}_I, \Omega_{I,k}, t-1) - \Lambda(\gamma_{\rm th}, t-1)\right]}{t!\varpi_k^{-1}\vartheta(L,l)^{-1}}. \quad (49)$$

The proof of Theorem 2 is concluded.

APPENDIX C
PROOF OF THEOREM 3

We consider the asymptotic expressions of OP in the interference-tolerant case, when $I_p \gg P_m$, and in the interference-limited case, when $P_m \gg I_p$. Making use of the fact that $\prod_{k=1}^{K-1}(1 - z_k) \approx 1 - \sum_{k=1}^{K-1} z_k$ for small z_k, the asymptotic expression of OP in (23) can be approximated as

$$P_{\rm out}^{\tt DHS} \approx F_{\gamma_1}(\gamma_{\rm th}) + \sum_{k=1}^{(K-1)/2} F_{\gamma_k}(\gamma_{\rm th}). \quad (50)$$

Consider the OP of DHS scheme for the interference-tolerant case, i.e., $I_p \gg P_m$. From (13), the transmit power is approximated by $\mathrm{P}_{k,i} \approx \sum_{m=1}^{M} \kappa P_m |g_{m,k,i}|^2$. Hence, the instantaneous SNRs for the first hop and the next kth dual-hop can be expressed, respectively, as

$$\gamma_1^T = \max_{j=1,\ldots,N_1} \sum_{m=1}^{M} \kappa\bar{\gamma}_P |g_{m,0,1}|^2 |h_{0,1,j}|^2, \quad (51)$$

$$\gamma_k^T = \max_{i=1,\ldots,N_{2k}} \min\Bigg(\sum_{m=1}^{M} \kappa\bar{\gamma}_P g_{m,2k-1,u^*} h_{2k-1,u^*,i}, \max_{j=1,\ldots,N_{2k+1}} \sum_{m=1}^{M} \kappa\bar{\gamma}_P g_{m,2k,i} h_{2k,i,j}\Bigg), \quad (52)$$

Similar to the approach developed for $F_{\gamma_1}(\gamma_{\rm th})$ in Theorem 1, the CDF of γ_1^T can be obtained as

$$F_{\gamma_1^T}(\gamma_{\rm th}) = 1 - \sum_{n=1}^{N_1} \frac{2\psi(N_1,n)\Xi(n\gamma_{\rm th}, \lambda_{D,1}, M)}{\Gamma(M)}, \quad (53)$$

Next, similar to the approach developed for $F_{\gamma_k}(\gamma_{\rm th})$ in Theorem 1, $F_{\gamma_k^T}(\gamma_{\rm th})$ can be obtained as

$$F_{\gamma_k^T}(\gamma_{\rm th}) = \prod_{i=1}^{N_{2k}} \Bigg[1 - \sum_{n=1}^{N_{2k}} \frac{4\Xi(\gamma_{\rm th}, \lambda_{D,2k-1}, M)}{\Gamma(M)^2} \times \psi(N_{2k},n)\Xi(n\gamma_{\rm th}, \lambda_{D,2k}, M)\Bigg]. \quad (54)$$

We now shift our focus to the derivation for the OP of DHS scheme in the interference-limited case, i.e., $I_p \ll P_m$. The transmit power can be approximated as $\mathrm{P}_{k,i} \approx \bar{\gamma}_I / \max_{l=1,\ldots,L} |\hat{f}_{k,i^*,l}|^2$, and the instantaneous SNRs for the first hop and the next kth dual-hop can be expressed, respectively, as

$$\gamma_1^L = \max_{j=1,\ldots,N_1} \left(\bar{\gamma}_I |h_{0,1,j}|^2 / \max_{l=1,\ldots,L} |\hat{f}_{0,1,l}|^2\right), \quad (55)$$

$$\gamma_k^L = \max_{i=1,\ldots,N_{2k}} \min\Big(\bar{\gamma}_I h_{2k-1,u^*,i} / \max_{l=1,\ldots,L} |\hat{f}_{2k-1,u^*,l}|^2, \max_{j=1,\ldots,N_{2k+1}} \bar{\gamma}_I |h_{2k,i,j}|^2 / \max_{l=1,\ldots,L} |\hat{f}_{2k,i,l}|^2\Big). \quad (56)$$

Recalling again Theorem 1, $F_{\gamma_1^L}(\gamma_{\rm th})$ can be obtain as

$$F_{\gamma_1^L}(\gamma_{\rm th}) = 1 - \sum_{l=0}^{L-1}\sum_{n=1}^{N_1} \frac{\vartheta(L,l)\psi(N_1,n)}{2\chi(\lambda_{D,1}, \gamma_{\rm th}, \Omega_{I,1})^{-1}}, \quad (57)$$

and the CDF of γ_k^L can be also obtained as

$$F_{\gamma_k^L}(\gamma_{\rm th}) = \prod_{i=1}^{N_{2k}} \Bigg[1 - \sum_{l=0}^{L-1} \frac{\vartheta(L,l)\chi(\lambda_{D,2k-1}, \gamma_{\rm th}, \Omega_{I,2k-1})}{4} \times \sum_{l=0}^{L-1}\sum_{n=1}^{N_{2k}} \frac{\vartheta(L,l)\psi(N_{2k},n)}{\chi(\lambda_{D,2k}, \gamma_{\rm th}, \Omega_{I,2k})^{-1}}\Bigg]. \quad (58)$$

Plugging $F_{\gamma_1^T}(\gamma_{\rm th})$ and $F_{\gamma_k^T}(\gamma_{\rm th})$ into (50) for the interference-tolerant case, and substituting $F_{\gamma_1^L}(\gamma_{\rm th})$ and $F_{\gamma_k^L}(\gamma_{\rm th})$ into (50) for the interference-limited case, we finally obtain the desired results as (14). The proof is concluded.

APPENDIX D
PROOF OF THEOREM 4

Similar to the proof of Theorem 3 in DHS scheme, the asymptotic expressions for the OP of BPS scheme can be rewritten as

$$P_{\rm out}^{\tt BPS} \approx \prod_{p=1}^{\Psi} \Big[\sum_{k=1}^{K-1} F_{\gamma_k}(\gamma_{\rm th})\Big]. \quad (59)$$

where $F_{\gamma_k}(\gamma_{\rm th})$ in the case $I_p \gg P_m$ can be obtained as

$$F_{\gamma_{1,k}^T}(\gamma_{\rm th}) = 1 - 2\Xi(\gamma_{\rm th}, \lambda_{D,k}, M)/\Gamma(M), \quad (60)$$

and that in the case $I_p \ll P_m$ can be obtained as

$$F_{\gamma_{1,k}^L}(\gamma_{\rm th}) = 1 - \sum_{l=0}^{L-1} \vartheta(L,l)\chi(\lambda_{D,k}, \gamma_{\rm th}, \Omega_{I,k})/2. \quad (61)$$

Finally, the proof of Theorem 4 is concluded.

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